THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — SCIENCE · CLASS 9 · SET 01

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — BIOLOGY — 16 questions · 32 marks

Q.1–7 MCQs (1 mark each); Q.8–9 Assertion–Reason. For Assertion–Reason: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

1. Cells of the pancreas secrete large amounts of digestive enzymes (proteins). Which pair of cell organelles will be especially abundant in these cells?

(a) Rough ER and Golgi apparatus (b) Smooth ER and lysosomes
(c) Plastids and vacuoles (d) Centrosome and chloroplasts

> **Ans: (a)** Proteins are made on ribosomes of RER and packaged/secreted by the Golgi apparatus.

2. Dry raisins are soaked in pure water for some hours and they swell. This is because water:

(a) moves out of the raisin by osmosis
(b) moves into the raisin through its semi-permeable membrane, since water is more concentrated outside
(c) moves into the raisin because sugar moves out by osmosis
(d) moves in both directions equally

> **Ans: (b)** Osmosis: net movement of water from higher water concentration (pure water) to lower (raisin) across a semi-permeable membrane.

3. Grass grows back quickly after it is mown. This is mainly due to the activity of:

(a) apical meristem at the stem tip
(b) lateral meristem
(c) intercalary meristem at the base of leaves
(d) cambium of the root

> **Ans: (c)** Intercalary meristem lies at the base of leaves/internodes and gives regrowth after cutting.

4. Which of the following statements about cardiac muscle is INCORRECT?

(a) Its action is involuntary.
(b) Its cells are cylindrical and branched.
(c) Its cells show light and dark striations.
(d) Its cells are long, unbranched and multinucleate.

> **Ans: (d)** Long, unbranched, multinucleate cells are characteristic of skeletal (striated) muscle, not cardiac muscle.

5. A solar panel of area 4 m^2 receives sunlight of insolation 0.8 kW m^{-2} for 5 hours. The solar energy received by the panel is:

(a) 3.2 kWh (b) 16 kWh (c) 4 kWh (d) 20 kWh

> **Ans: (b)** Energy = 0.8 × 4 × 5 = 16 kWh.

6. A single-celled organism has no membrane-bound nucleus or organelles but has a cell wall. It is placed in the kingdom:

(a) Protista (b) Fungi (c) Monera (d) Plantae

> **Ans: (c)** Prokaryotes (bacteria) belong to kingdom Monera.

7. Sugarcane is propagated by planting pieces of its stem. This method of reproduction is called:

(a) sexual reproduction (b) vegetative propagation
(c) budding (d) spore formation

> **Ans: (b)** New plants arise from vegetative parts (stem cuttings) – asexual vegetative propagation.

8. **Assertion (A):** Plasmolysis occurs when a plant cell is placed in a hypertonic solution.
Reason (R): Water moves out of the cell by osmosis and the protoplast shrinks away from the cell wall.

> **Ans: (a)** Hypertonic outside → water leaves the cell → protoplast shrinks; R correctly explains A.

9. **Assertion (A):** Ozone depletion allows more ultraviolet radiation to reach the Earth's surface.
Reason (R): Ozone is formed only in the troposphere, close to the ground.

> **Ans: (c)** A is true. R is false: most ozone is in the stratosphere (the ozone layer).

10. Define budding. Name one organism that reproduces by budding.

> Budding: a small outgrowth (bud) develops on the parent body, grows, and detaches to become a new individual. [1] Example: *Hydra* / yeast. [1]

OR

What is fragmentation? Give one example.

> Breaking of the body into fragments, each of which grows into a complete new individual. [1] Example: *Spirogyra*. [1]

11. Name the type of tissue present in: (i) the hard shell of a walnut, (ii) the pulp of a ripe tomato, (iii) the lining of kidney tubules.

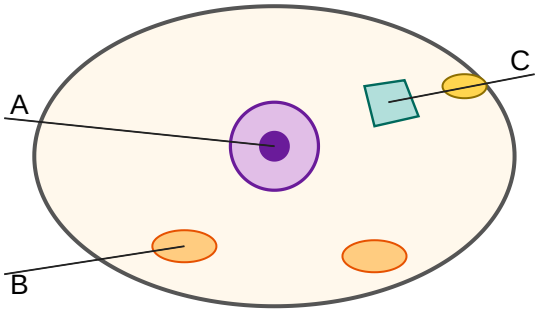
> (i) Sclerenchyma [1] (ii) Parenchyma [1] (iii) Cuboidal epithelium [1]

OR

Name the muscle tissue found in: (i) the wall of the alimentary canal, (ii) the heart, (iii) the biceps attached to the bone. State one feature of each.

> (i) Smooth (unstriated) – spindle-shaped, involuntary [1] (ii) Cardiac – branched, striated, involuntary [1] (iii) Skeletal – long, striated, multinucleate, voluntary [1]

12. The figure shows an animal cell with three labelled parts A, B and C. Identify each and write one function of each.



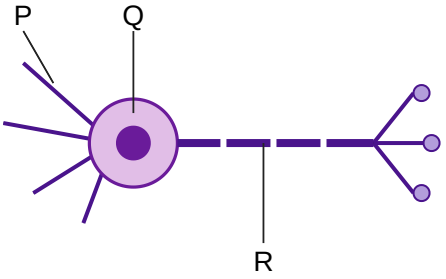
> A – Nucleus: controls cell activities/stores DNA. [1] B – Mitochondrion: releases energy as ATP (powerhouse). [1] C – Golgi apparatus: modifies, packages and dispatches materials. [1]

13. Copy and complete the table by writing 'Present' or 'Absent' for each cell:

Structure	Human cheek cell	Spinach leaf cell	Bacterium
Cell wall			
Membrane-bound nucleus			
Chloroplast			

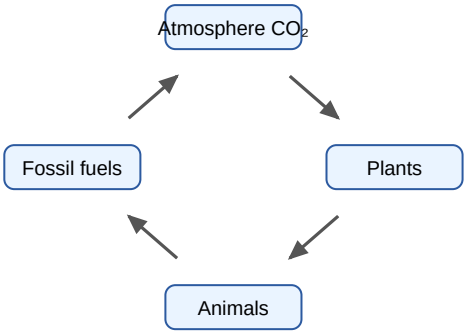
Cell wall: Absent, Present, Present [1]; Membrane-bound nucleus: Present, Present, Absent [1]; Chloroplast: Absent, Present, Absent [1]

14. The figure shows a nerve cell (neuron) with parts P, Q and R labelled. Name P, Q and R and state in which direction the nerve impulse travels through the neuron.



P – Dendrites (receive impulse); Q – Cell body/cyton; R – Axon (conducts impulse away from the cell body). [½ each name + function = 2] Impulse travels dendrites → cell body → axon → nerve endings. [1]

15. A forest research station studied how carbon moves in nature. Students drew the cycle shown below. They noted that the amount of CO_2 in the air has been rising since the burning of coal and petroleum increased, and that large areas of forest are being cleared for roads and farms.



(a) Name the process by which green plants remove CO_2 from the atmosphere.

Photosynthesis. [1]

(b) Name one human activity, other than deforestation, that increases atmospheric CO_2.

Burning of fossil fuels (coal/petroleum) / vehicular emissions / industries. [1]

(c) Explain how deforestation can intensify the greenhouse effect.

Fewer trees → less CO_2 removed by photosynthesis [1]; burning/decay of cut trees releases stored carbon as CO_2 → more heat trapped. [1]

OR

(c) What is the greenhouse effect? Why is its enhancement harmful?

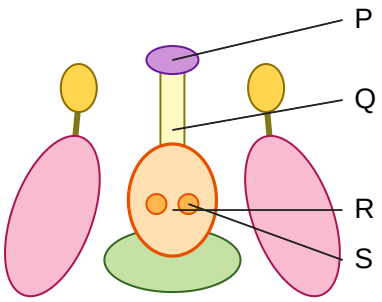
Natural trapping of outgoing infrared heat by gases like CO_2, CH_4, water vapour keeps Earth warm. [1] Excess gases trap more heat → global warming, melting ice, climate change. [1]

16. (i) What are lysosomes? Why are they called ‘suicide bags’ of the cell?
(ii) State any three differences between prokaryotic and eukaryotic cells.

(i) Membrane-bound sacs of digestive enzymes [1]; on cell damage/ageing they burst and the enzymes digest the cell itself [1]. (ii) Any 3: nucleus (no true nucleus/true nucleus), nuclear membrane, membrane-bound organelles, DNA arrangement, size, example [3 × 1]

OR

Study the diagram of a flower. (i) Identify parts P, Q, R and S. (ii) Name the process by which pollen reaches the stigma. (iii) What do the ovule and the ovary develop into after fertilisation?



(i) P – Stigma, Q – Style, R – Ovary, S – Ovule [4 × ½ = 2] (ii) Pollination [1] (iii) Ovule → seed [1]; ovary → fruit [1]

SECTION B — CHEMISTRY — 11 questions · 25 marks

Q.17–19 MCQs; Q.20 Assertion–Reason (options as in Section A).

17. A beam of light is passed through four liquids kept in transparent beakers. The path of the beam will be clearly visible in:

(a) a salt solution (b) a glucose solution
(c) a starch solution (d) a copper sulphate solution

Ans: (c) Starch solution is a colloid – its particles scatter light (Tyndall effect).

18. Whether water is obtained from a river, a well or by burning hydrogen in oxygen, hydrogen and oxygen are always found in the mass ratio 1 : 8. This illustrates the law of:

(a) conservation of mass (b) constant proportions
(c) gaseous volumes (d) radioactive decay

Ans: (b) Constant proportions: a pure compound always has elements in a fixed mass ratio.

19. An ion has 13 protons and 10 electrons. It is represented as:

(a) Al^{3+} (b) Al^{+} (c) Na^{+} (d) Mg^{2+}

Ans: (a) Charge = +13 − 10 = +3; the element with Z = 13 is aluminium.

20. **Assertion (A):** Isotopes of chlorine, ^{35}Cl and ^{37}Cl, have identical chemical properties.
Reason (R): Isotopes of an element have different numbers of protons.

Ans: (c) A is true (same electrons). R is false: isotopes have the same number of protons but different numbers of neutrons.

21. The thickness of a hair is 0.00008 m. Express it (i) in scientific notation and (ii) in micrometres (1 µm = 10^{-6} m).

(i) 8×10^{-5} m [1] (ii) $8 \times 10^{-5} \div 10^{-6} = 80$ µm [1]

22. Name the technique used to separate (i) the different dyes present in black ink and (ii) cream from milk. State the principle behind each.

(i) Chromatography – different solubility/adsorption of dyes in the solvent [1] (ii) Centrifugation – difference in density when spun [1]

23. Calculate the molecular mass of carbon dioxide (C = 12 u, O = 16 u) and the ratio by mass of carbon to oxygen in it.

$CO_2 = 12 + 2 \times 16 = 44$ u [1]; C : O = 12 : 32 = 3 : 8 [1]

24. A saline solution is prepared by dissolving 18 g of sodium chloride in 282 g of water. (i) Calculate the mass percentage (% m/m) of the solution. (ii) How much sodium chloride is present in 500 g of this

solution?

(i) Mass of solution = 18 + 282 = 300 g [½]; % m/m = (18/300) × 100 = 6 % [1½] (ii) 6 % of 500 g = 30 g [1]

25. State any three postulates of Bohr's model of the atom.

(i) Electrons revolve in fixed circular orbits (shells) without radiating energy. (ii) Each orbit has a fixed energy; K, L, M… (iii) Energy is absorbed/emitted only when an electron jumps between orbits. (iv) Maximum electrons $2n^2$. [Any 3 × 1]

OR

Differentiate between isotopes and isobars with one example of each.

Isotopes: same atomic number, different mass number (e.g. ^{12}C, ^{14}C) [1½]. Isobars: same mass number, different atomic number (e.g. ^{40}Ar, ^{40}Ca) [1½].

26. A teacher tested three unlabelled samples P, Q and R, and recorded the observations in the table.

Sample	Appearance	Light path visible?	Settles on standing?	Residue on filter paper?
P	Clear	No	No	No
Q	Cloudy	Yes	Yes	Yes
R	Translucent	Yes	No	No

(a) Classify the samples P, Q and R as solution, suspension or colloid.

P – solution; Q – suspension; R – colloid. [1]

(b) Why can sample R not be separated by ordinary filtration even though it is not a true solution?

Colloidal particles are very small (1–100 nm) and pass through filter paper. [1]

(c) Name the process to obtain the dissolved solute from sample P and write one example of sample P.

Evaporation/crystallisation [1]; example – salt/sugar solution [1]

OR

(c) Arrange P, Q and R in increasing order of particle size and justify using the observations.

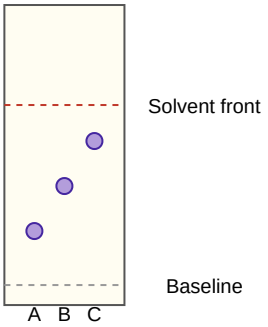
P < R < Q [1]; P shows no scattering and passes through filter; R scatters but passes filter; Q scatters, settles and is retained. [1]

27. A mixture contains common salt, sand and iron filings. Describe, step by step, how you will separate the three components, naming the principle used at each step.

1. Magnet removes iron filings (magnetic property) [1½] 2. Add water, stir – salt dissolves, sand does not [1] 3. Filter – sand is the residue [1] 4. Evaporate/crystallise the filtrate – salt obtained [1½]

OR

The chromatogram of a dye mixture is shown (solvent front is 100 mm above the baseline). (a) State the principle of paper chromatography. (b) Find the R_f value of spots A, B and C. (c) Which component is the most soluble in the solvent?



(a) Components of a mixture move with the solvent at different rates according to their solubility in the solvent and adsorption on paper. [2] (b) R_f = distance moved by spot ÷ distance moved by solvent: A = 30/100 = 0.30; B = 55/100 = 0.55; C = 80/100 = 0.80 [2] (c) C – it travelled farthest. [1]

SECTION C — PHYSICS — 12 questions · 23 marks

Q.28–33 MCQs; Q.34 Assertion–Reason (options as in Section A).

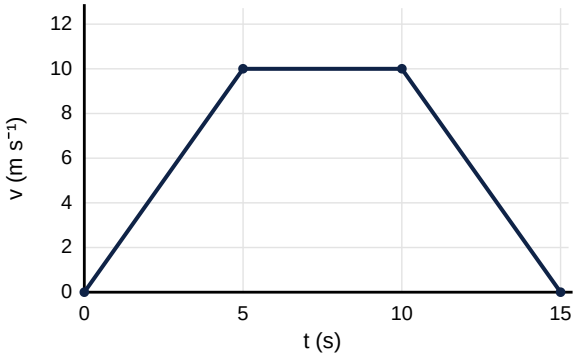
28. A runner completes one full lap of a 400 m circular track in 80 s. Her average speed and average velocity for the lap are respectively:

(a) 5 m s^{-1}, 0 (b) 0, 5 m s^{-1}

(c) 5 m s^{-1}, 5 m s^{-1} (d) 0, 0

Ans: (a) Average speed = 400/80 = 5 m s^{-1}; displacement is zero, so average velocity is zero.

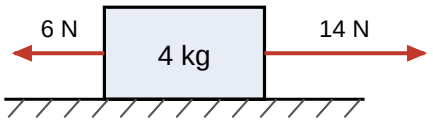
29. The velocity–time graph of a car is shown. The distance covered by the car in the first 10 s is:



(a) 50 m (b) 75 m (c) 100 m (d) 125 m

Ans: (b) Area = ½ × 5 × 10 + 5 × 10 = 25 + 50 = 75 m.

30. The figure shows the horizontal forces acting on a 4 kg block on a smooth surface. Its acceleration is:



(a) 5 m s^{-2} (b) 2 m s^{-2} (c) 0.5 m s^{-2} (d) 8 m s^{-2}

Ans: (b) Net force = 14 − 6 = 8 N; a = F/m = 8/4 = 2 m s^{-2}.

31. A rocket moves upward by throwing hot gases downward. This is an example of Newton's:

(a) first law (b) second law

(c) third law (d) law of gravitation

Ans: (c) Action (gases pushed down) and reaction (rocket pushed up) – third law.

32. A 2 kg body is raised through a height of 5 m. The work done against gravity is ($g = 10\ m\ s^{-2}$):

(a) 10 J (b) 50 J (c) 100 J (d) 20 J

> **Ans: (c)** W = mgh = 2 × 10 × 5 = 100 J.

33. A sound wave has a frequency of 250 Hz and a wavelength of 1.36 m. The speed of the wave is:

(a) 183.8 m s^{-1} (b) 1.36 m s^{-1} (c) 250 m s^{-1} (d) 340 m s^{-1}

> **Ans: (d)** $v = f\lambda = 250 \times 1.36 = 340\ m\ s^{-1}$.

34. **Assertion (A):** A force can change the direction of a moving body.
Reason (R): A force can change the speed of a body.

> **Ans: (b)** Both statements are true, but changing speed does not explain changing direction.

35. A car moving at 10 m s^{-1} accelerates uniformly at 2 m s^{-2} for 5 s. Find (i) its final velocity and (ii) the distance covered in this time.

> (i) $v = u + at = 10 + 2\times5 = 20\ m\ s^{-1}$ [1] (ii) $s = ut + \frac{1}{2}at^2 = 10\times5 + \frac{1}{2}\times2\times25 = 75$ m [1]

36. A cyclist stops pedalling but the bicycle keeps moving for some distance. Name the law that explains this and state why the cycle finally stops.

> Newton's first law (inertia) [1]; friction/air resistance (unbalanced force) slows it. [1]

OR

A gun recoils backward when a bullet is fired. Name the law that explains this and state why the recoil velocity is much smaller than the velocity of the bullet.

> Newton's third law/conservation of momentum [1]; gun has a much larger mass than the bullet. [1]

37. A pump raises 600 kg of water to a height of 20 m in 2 minutes. Calculate (i) the work done and (ii) the power of the pump ($g = 10\ m\ s^{-2}$).

> (i) W = mgh = 600 × 10 × 20 = 1,20,000 J [1½] (ii) P = W/t = 120000/120 = 1000 W [1½]

38. During a hockey match, Seema, the goalkeeper, stops a ball of mass 0.16 kg moving at 25 m s^{-1}. The ball comes to rest in 0.05 s when she stops it with her padded gloves.

(a) Calculate the momentum of the ball before it is stopped.

> $p = mv = 0.16 \times 25 = 4\ kg\ m\ s^{-1}$. [1]

(b) Find the average force exerted by the ball on her gloves.

> $F = \Delta p/t = 4/0.05 = 80$ N. [1]

(c) Why do goalkeepers pull their hands backward while stopping a fast ball?

> Increases the time of stopping [1]; so rate of change of momentum (force) decreases and injury is avoided. [1]

OR

(c) If the ball were stopped in 0.2 s instead, what force would be exerted? Compare it with the first case.

> F = 4/0.2 = 20 N [1]; it is 4 times smaller than 80 N since time is 4 times longer. [1]

39. A cyclist rides 6 km east in 20 minutes and then 8 km north in the next 20 minutes. Calculate (i) the total distance covered, (ii) the magnitude of displacement, (iii) the average speed and (iv) the average velocity. Why is the average speed greater than the average velocity?

> (i) 14 km [1] (ii) $\sqrt{(6^2+8^2)} = 10$ km [1] (iii) $14 \div (40/60) = 21\ km\ h^{-1}$ [1] (iv) $10 \div (2/3) = 15\ km\ h^{-1}$ [1] Because distance (path length) is greater than displacement. [1]

O R

A body starts from rest, accelerates uniformly at 2 m s^{-2} for 10 s, moves with constant velocity for the next 20 s and then comes to rest uniformly in 5 s. Find (i) the maximum velocity, (ii) the distance in the first 10 s, (iii) the distance at constant velocity, (iv) the retardation and (v) the total distance.

> (i) 20 m s^{-1} [1] (ii) ½×2×100 = 100 m [1] (iii) 20×20 = 400 m [1] (iv) 20/5 = 4 m s^{-2} [1] (v) 100 + 400 + ½×20×5 (=50) = 550 m [1]